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ENGS 23 Final Report — Duck Car

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1. Introduction:

The goal of this project is to design a closed loop system that upholds stability and precision.

This means understanding different aspects of the closed loop system. For the Duck Car, we had to characterize the motor attached to the car as well as understand and calibrate the sensor that proved the feedback of the system. Once understanding other elements of the closed loop system we had to design a controller in order to stabilize the system and produce zero steady state error.

For the Duck Car, this means stopping the car at a steady state value, distance from object in front of it, as well as to not allow any peak overshoot where the car would get too close to the wall and then have to back up to restabilize at steady state. Our goal was to make this system as efficient as possible by allowing the response to be reached quickly and be well damped. We attempted to maximize the closed loop bandwidth while limiting the magnitude of the frequency response, and output to below 1 DB above the DC value and 12 Volts respectively. In addition the system must be able to respond to any input disturbance and minimize its effect on the steady state. In order to achieve this after analyzing the uncompensated system, we designed and implemented a Proportional Derivative compensator.

2. System Modeling:

Sensor Calibration:

The duck car uses a sensor to detect how far it is from a particular surface. By using this information and a compensator in negative feedback, we can control how far the car goes before stopping in front of a wall. The first step in determining the design of our compensator was calibrating the sensor in the car. By measuring the voltage output from the sensor when placing a sheet of paper at different distances from it, we were able to find the general shape of the motor's output. For the MATLAB code and graph of sensor calibration see [Figure 1](#) in the appendix.

Motor Characterization:

To understand the dynamic behavior of the car's motor, we input a step input which impacts the motor's operation. As this step input was applied, we monitored and recorded the current flowing into the motor. This data provided us with a clear representation of the motor's response to the input. Knowing that the motor can be approximated as a first-order system, we analyzed the recorded information to extract the time constant. Knowing this value allowed us to characterize the motor's behavior. From the scope image provided below([Figure 2](#)) the time constant was observed to be 200 ms. This information, in turn, facilitated the development of a transfer function model for the motor. This model enabled us to predict its response to different inputs, contributing significantly to the optimization of its performance within the car's control system.

Along with the pole of the transfer function. From the lecture it is known that there is an integrating component within the motor therefore requires an additional pole with location at 0 to be accounted for when coming up with the whole motor transfer function.

In addition to finding the transfer function of the motor it is necessary to find the gain of it as well. This was achieved by measuring the voltage of the blue wire which is the input of the motor car, knowing V_{Ref} , and deriving the final gain value through math. The value that we want to observe is what's coming out of the "Motor Car" block. Since the transfer function of the sensor is known, the output of the "Motor Car" block, D , can be derived from an equation. Through the following math we were able to derive the gain.

Measured Values:

$$V_{Ref} = 1.5003 \text{ V} \ \& \ V_{Blue \ Wire} = 1.4388$$

Deriving Gain of Motor Block (D):

$$D = \frac{0.15}{V_{Ref} - V_{Blue \ Wire}} = \frac{0.15}{1.5003 - 1.4388} = 2.439$$

Transfer Functions:

$$TF_{Motor \ Car} : MotorOut = \frac{2.439}{s(s + 5)} MotorIn$$

$$TF_{Sensor} : V_{sensor} = 0.15 \frac{V}{D} \cdot Distance$$

To see the total block diagram of the system see [Figure 3](#) in the appendix below.

2. Analysis of Stability for Uncompensated System:

Key Attributes From Figure 4:

Rise Time = 4.08 Seconds Settling Time = 7.39 Seconds

Pole Locations: $s = 0$, $s = -5$

From initial observations of the Root Locus and Step Response plots it's clear to see that an uncompensated system that consists of the motor and sensor is stable. However, the rise time is slow resulting in the car moving slowly until it reaches an error of zero. This means that the distance the car is from what's in front of it is equal to the distance sent in terms of voltage from the potentiometer.

When testing the car in the real world, compared to only simulation from Sisotool, it was moving at a slower speed up until it reached a distance that was specified with the potentiometer's voltage. Based on this behavior we made it a specific goal to have the car approach the wall, or specific target faster, while still maintaining steady-state error to be zero.

3. Design Specifications:

When creating the system design, our main objective was to achieve a faster rise time, which is correlated to the speed of the car. A fast rise time is crucial for enhancing the system's responsiveness and ensuring proper adjustment to dynamic changes, which can include the wall where the sensor is reflective off to move closer or farther. At the same time, we wanted to mitigate peak overshoot to the minimum possible extent. This approach is to prevent deviations from the reference distance set by the potentiometer, therefore minimizing steady state error. By

prioritizing both a fast rise time and minimal peak overshoot, ensuring a precise and efficient control mechanism for the vehicle.

Specifically we would want to reach these specific design specifications:

Damped System:

Rise Time < 4 Seconds.

Settling Time < 10 Seconds

Peak Overshoot ≈ 0

4. Compensator Design:

We opted for a Proportional-Derivative controller for its swift response to system changes and minimized oscillations. These attributes align with our system control specifications and goals. According to the lecture notes a PD controller consists of a gain along with a specific pole and zero. Using the Sisotool Feature in MATLAB we were able to analyze the Step Response and Root Locus of our system to build a compensator that meets our design specifications.

First, in order to make the compensator buildable we needed to ensure that the pole's location was greater relative to the zero's location. Once we placed our pole and zero we could tweak their specific locations and observe what would happen to the Step Response.

When only introducing a pole this would result in the peak overshoot to be larger than our specifications. Therefore, we need to introduce a real zero in order to make the step response overdamped again to eliminate oscillation. Once both the zero and the pole were added to the

Root Locus we tweaked the gain in order to change the rise time to be shorter in order to allow the car to move faster as it approached the specified distance of an obstacle in front of it.

To see a graphical representation of what happens to the Root Locus when introducing a real pole and real zero see [Figure 5](#) for more details. This enforces how the behavior of the plot changes as described above.

Final Compensator Transfer Function:

$$G_c = 1.25 \frac{1 + 3.7s}{1 + 1.9s}$$

Important Values:

Gain = 1.25

Location of Zero: $s = -0.2703$

Location of Pole: $s = 0.5263$

For more information about the Root Locus and the Step Input Response Graph see [Figure 6](#) in the appendix.

5. Compensator Implementation:

According to the lecture and lab notes a PD Controller can be designed with three different resistor values R_1 , R_2 , and R_{stop} along with 1 Capacitor. There is a standard topology used to implement this type of controller (See [Figure 7](#)). The transfer function for this type of controller

is given by $-K_p \frac{(Td+\alpha)s + \alpha Td}{Td(s+\alpha)}$, where $K_p = \frac{R_2}{R_1}$, $Td = \frac{1}{R_1 C_1}$, $\alpha = \frac{1}{R_s C_1}$. In terms of the

resistor and capacitor values, the transfer function is given as $\frac{-R_2}{R_1} \left(\frac{R_1 C_1 s + R_s C_1 s + 1}{R_s C_1 s + 1} \right)$. To ensure that our compensator transfer function was the same as our design, we had to make sure that $R_1 C_1 + R_s C_1 = 3.7$, and that $R_s C_1 = 1.9$. To meet these specifications we chose the following values (See [Figure 8](#) for algebra):

$$R_1 = 80k\Omega, R_2 = 100k\Omega, R_s = 190k\Omega, C_1 = 10\mu F$$

6. Experimental Procedure: (Collin)

We set out to test our car after solving for theoretical values to observe how they translated to the implementation of our system for each system design. We first tested our uncompensated car to visually see how the car would respond with an uncompensated system. In this case we observed that the car reached steady state very slowly and was clearly not very efficient.

Secondly, we developed a compensated system using a proportional controller. The compensated system with a proportional controller during testing showed a fast rising time due to our large gain but had high steady state error and peak overshoot. The Proportional controller allowed us to obtain a stable system that reached steady state efficiently, however due to our specifications we were unable to use this system due to the steady state error. Thirdly, we developed a system using a PD controller as our compensator with a gain of three. We observed that the car had an efficient rising time which resulted in a small peak overshoot. Even though our theoretical response in MATLAB displayed no steady state error, we still observed some

peak overshoot during our physical testing and evaluation of this system. Next, In order to meet our specifications to observe zero steady state error we decrease the gain of our compensator equal to one. During this test, the car did not experience any peak overshoot or steady state error, however, the car didn't approach steady state fast enough and it wasn't efficient enough. Lastly, In order to optimize the rise time to steady state and experience no steady state error, we re-increased our gain to 1.25. When we implemented this system, we observed an efficient and fast rise time to reach steady state and no steady state error or peak overshoot. Therefore achieving our experimental specifications.

To test our system after each design, we would initially place our car about 30 inches away from a foam barrier and turn our car on to observe how it would approach the steady state distance of four inches. Next we would place our car at the steady state value of four inches away from the foam board and observe how our car would react to moving the foam board forward and backward randomly. Again we would pay close attention to peak overshoot.

7. Appendix:

Figure 1: MATLAB Calibration

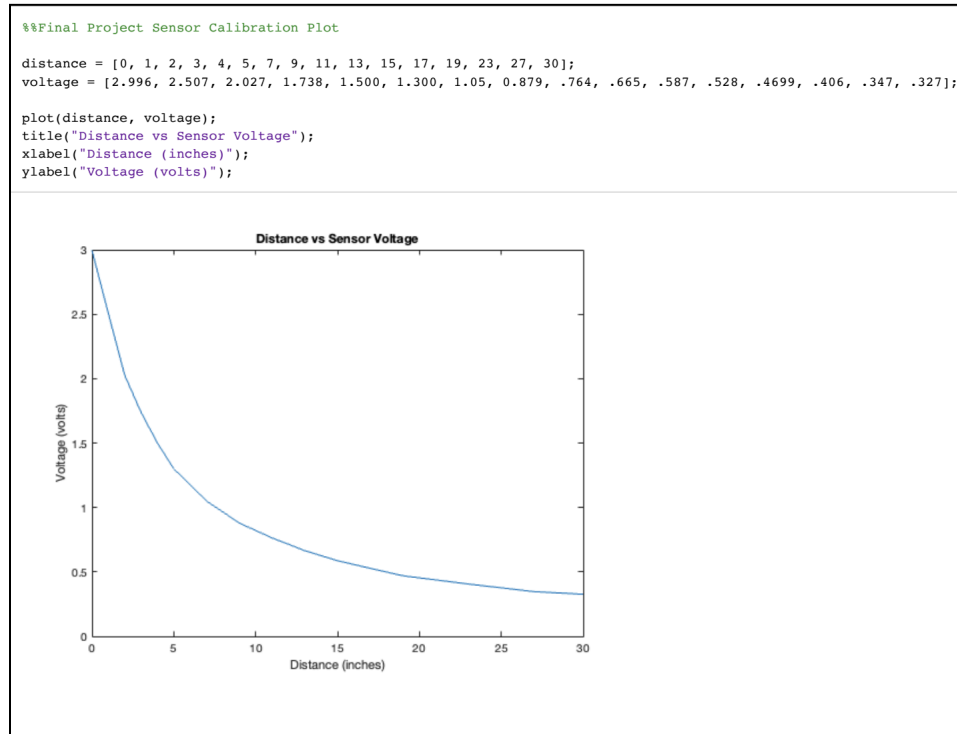


Figure 2: Given a Step Input (Yellow) Measured Current Behavior (Pink)

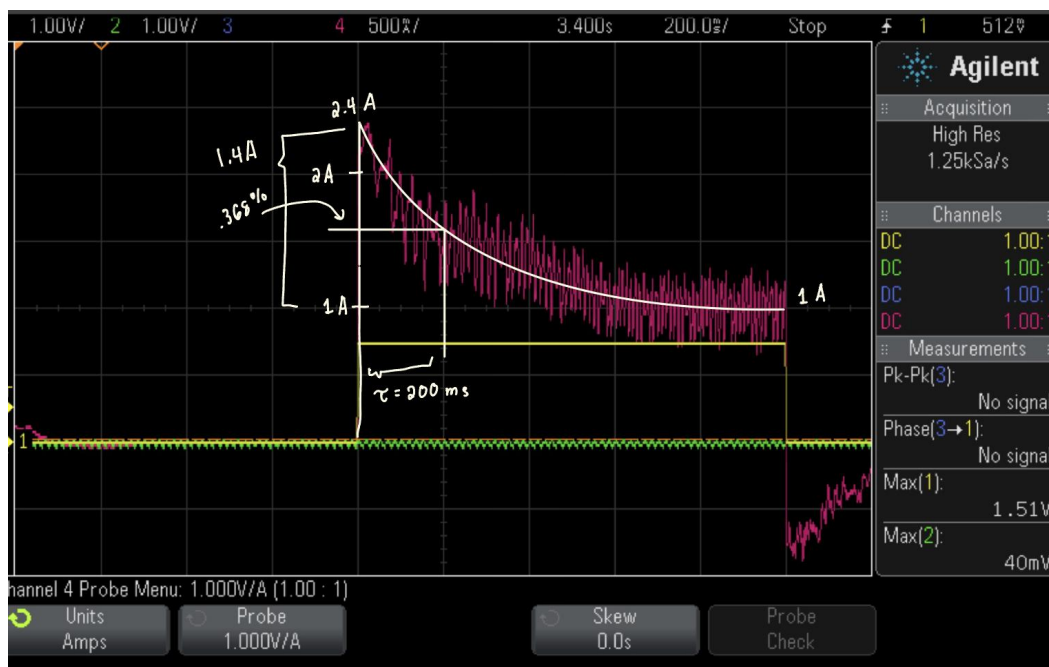


Figure 3: Total Block Diagram For System

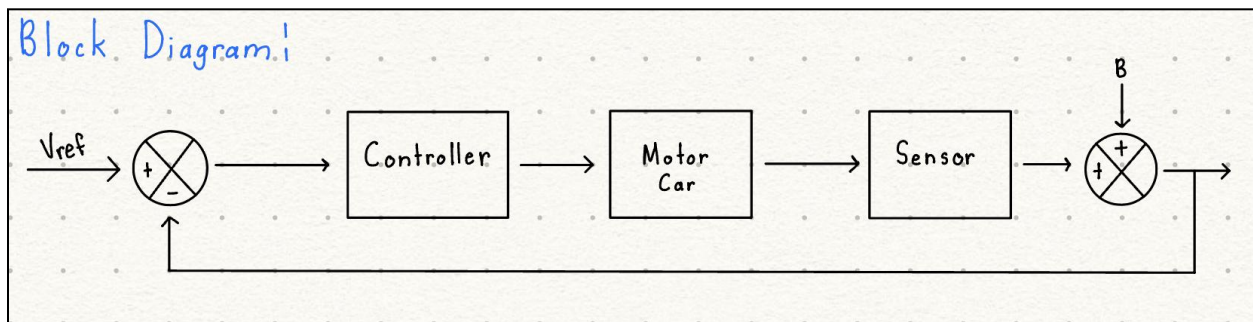


Figure 4: Root Locus and Step Response Graph of Uncompensated System

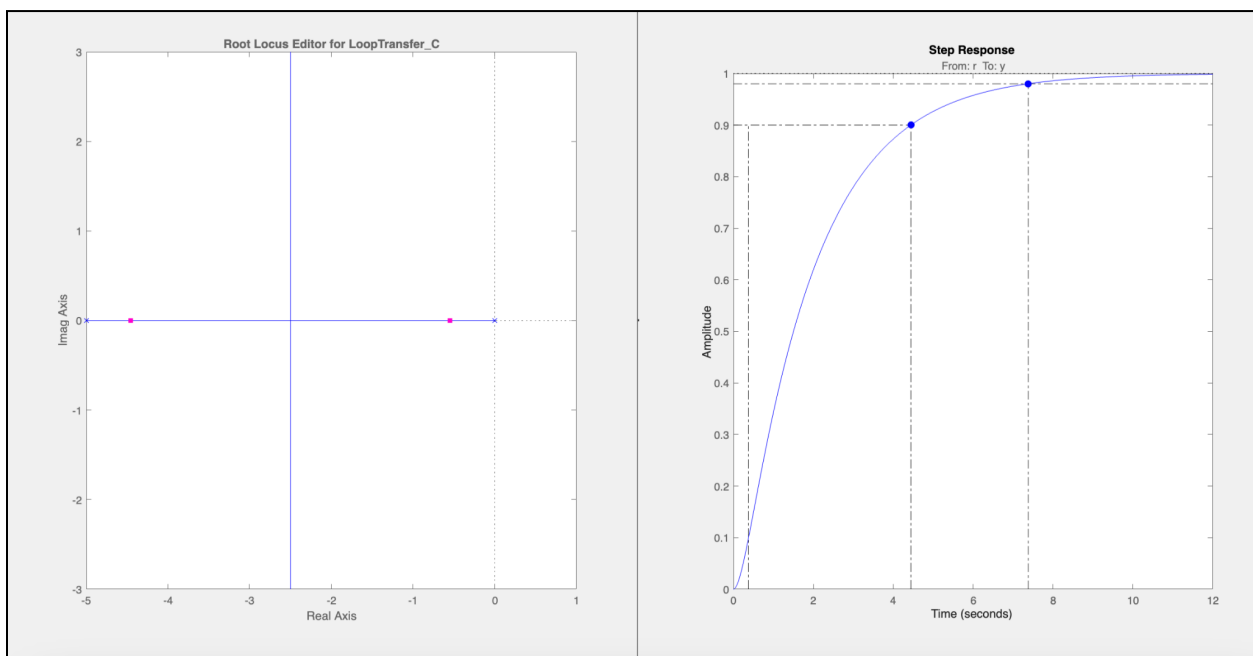


Figure 5: The Step Response Graph is the result of only having a pole. The Step Response Graph on the right is the result of having a pole and the zero.

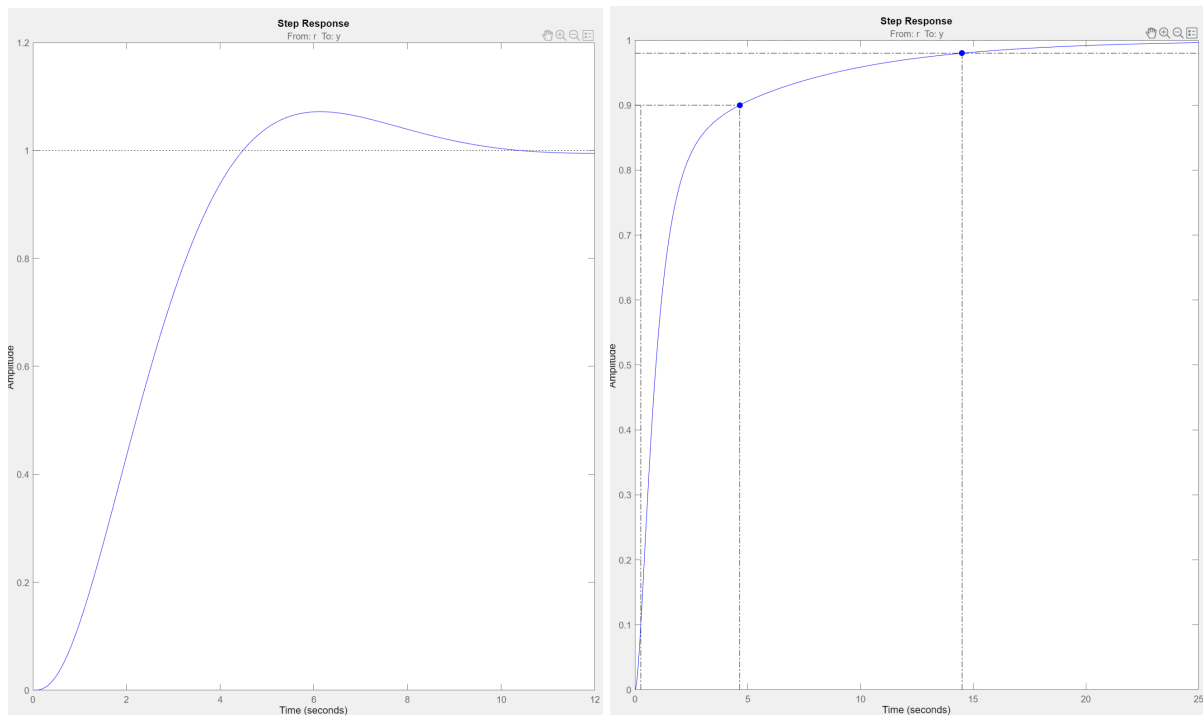


Figure 6: The Root Locus Graph and Step Response Graph of our final compensator.

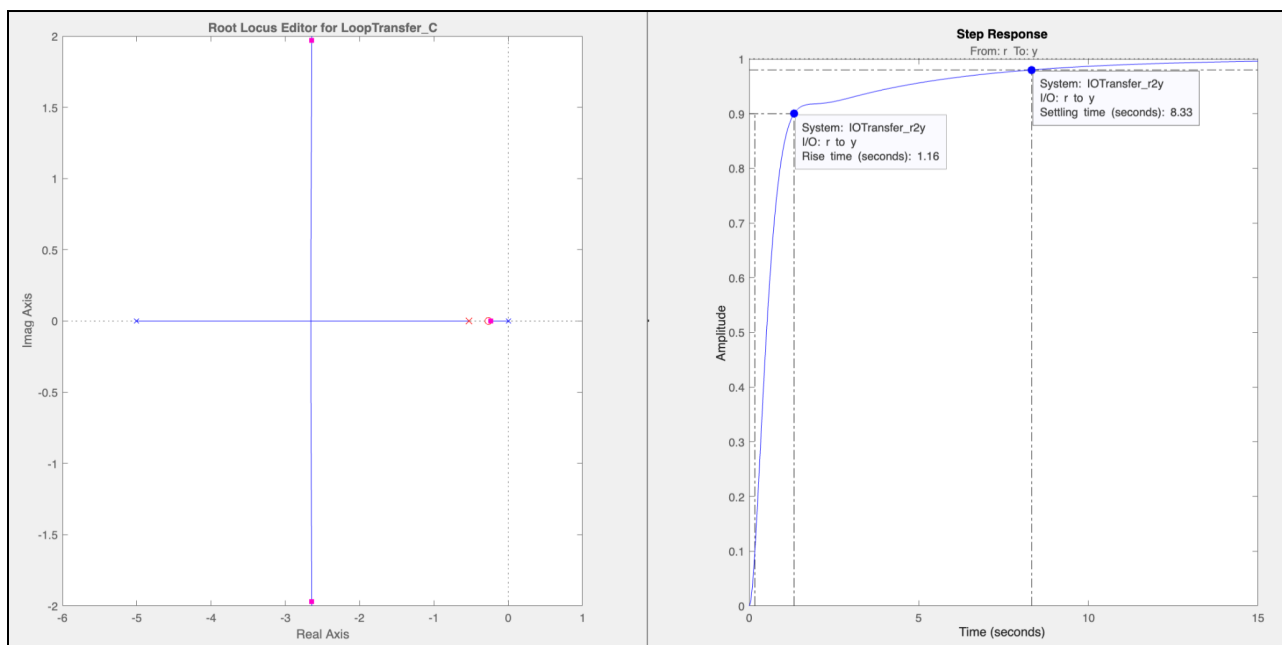


Figure 7: Final Compensator Topology

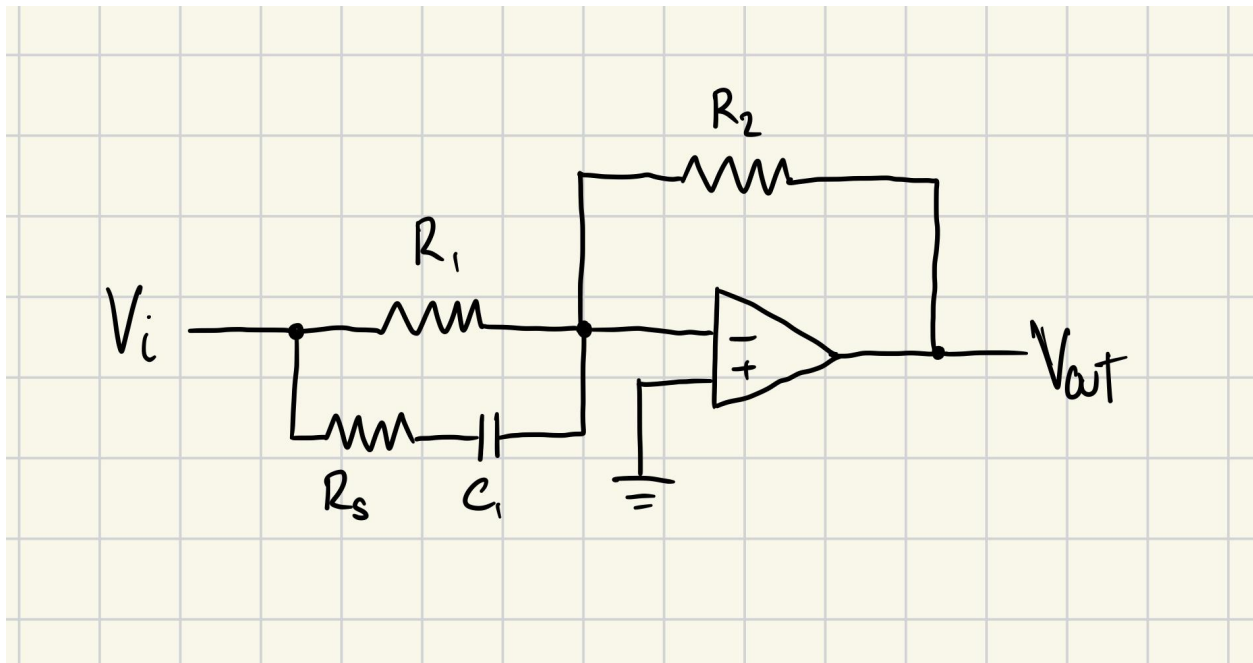


Figure 8: Compensator Component Value Algebra

